ActInf GuestStream 048.1 ~ Arthur Juliani & Adam Safron "Deep CANALs"

GuestStream | Arthur Juliani & Adam Safron | 1:52:12

Hello and welcome everyone. It's Active Inference Guest Stream number 48.1 on July 19th, 2023.
We're here with Arthur Giuliani and Adam Saffron, also joined by Michael Edward Johnson in
discussion. We're going to have a presentation and then a conversation about Deep Canals,
a Deep Learning Approach to Refining the Canalization Theory of Psychopathology.
So thank you all for joining and looking forward to this presentation and discussion.
Thanks, Daniel. I'm very happy to be here with Adam and Mike to talk about this work and to discuss
more broadly this general direction that's emerged over the past few years to try to
utilize some of the concepts from the active inference world and computational modeling
more generally to understand what's going on both in psychopathology and in psychedelic therapy.
Yeah, so I'll give a brief presentation and we'll move into a kind of more general discussion of this
work. It'll start with some kind of general considerations around psychedelic therapy,
some of the neuroscience of psychedelics, and then move kind of more specifically into the model.
So there is this class of drugs, the serotonergic psychedelic drugs,
which include psilocybin, DMT, LSD, and others. And the past decade has seen a resurgence of interest in
psychedelic therapy, psychedelic assisted psychotherapy, where one or multiple of these drugs is used in
conjunction with a kind of psychotherapeutic protocol. And there's been a lot of investment of money and
time and clinical resources into utilizing these drugs for treatments, depression, anxiety disorders,
substance use disorders, and others. And there seems to be quite a lot of positive signals suggesting that
these kinds of therapeutic interventions can actually be quite effective. So there's a number of studies
suggesting that psilocybin assisted psychotherapy can be helpful for depression, anxiety, substance use disorder,
and others. And as this work advances, there are more and more larger clinical trials comparing it, for example, to
the standard treatments with SSRIs, with multiple studies suggesting that psilocybin assisted psychotherapy is
at least as effective as the Lexapro kind of then care.
All of this seems to suggest, and the timelines are pointing at this idea that
psilocybin treatment for major depressive disorder may be only a couple of years away
of becoming approved as an actual treatment in the United States, which means that it's quite important
for us to understand what's going on mechanistically in the brain so that these kinds of treatments can be
best optimized, best determined how to be used,
for which kinds of mental issues people might be having, and under what kinds of treatment protocols
we might expect the most success. I think this will also ultimately hopefully lead to the development
of even better drugs, even better treatment protocols,
of the treatment protocols, and much more personalized care.

So some of the basics of how these drugs work on the brain, for those who are unfamiliar, right, so these are the serotonergic psychedelics. They are serotonergic because they act as agonists for a number of serotonin receptors in the brain and the central nervous system more broadly.
So it's thought that most specifically these drugs like psilocybin work through agonizing the serotonin 5-HT2A receptor, and this has been demonstrated multiple times by using blockers, antagonists of the 5-HT2A receptor,
which seem to largely remove the effect of psilocybin or other psychedelic drugs.
And what this 2-HT2A receptor does when it's agonized is it excites the neuro, making it more likely to fire.
And this is kind of essential to some of the downstream effects.
So given that it seems that this is this 2-HT2A receptor that psychedelics are mainly targeting, the question then becomes, well, where in the brain are these 2-HT2A receptors most widely expressed?
And research has focused on, I would say, three kind of key areas over the past decade or so. The first being the thalamus, which is a region involved in the gating of sensory information into the cortex.
There's a lot of expression in the cortex itself. Expression, for example, in the visual system, which is perhaps not surprising given the prevalence of visual hallucinations and technologies. But therapeutically, a lot of focus has been on the prefrontal cortex, where there is also a dense expression of these receptors. And then also more recent work has looked at the cloustrm, which is a small brain region responsible for coordinating the transition dynamics of cortical activation. And this, too, is likely of essential importance to the psychedelic effect.
So kind of moving up one level of abstraction to fMRI studies, there are a few kind of like key takeaways which have been identified from a few different fMRI studies looking at psychedelic use.
Kind of the main theme has largely been this idea of increased entropy or increased complexity of the cortical activation in the brain. So there are various measures of this. On the right here, there's a figure of one of the common measures of complexity and looking at various different kinds of states that an individual might be in and how that relates to complexity. So you have individuals in vegetative states or anesthetized states having relatively low brain activity complexity, you have individuals in awake states having higher complexity, and then individuals who have ingested either psilocybin or LSD showing the highest brain complexity. Now, along with this, rather than just this kind of like single unified score, there's work looking particularly at the changes in functional connectivity between different brain regions or different brain networks. And the general trend seems to be that you have decreased within network functional connectivity and increase between network functional
connectivity, right, which this kind of can be interpreted as different brain networks, which typically don't communicate as much with each other, more likely to communicate with each other under the effects of psychedelics. And the kind of like internal coherence of networks is reduced under psychedelics.
People have maybe heard of this default mode network, right, which is responsible for a lot of self-referential processing, which is responsible, which is typically activated, for example, when individuals are

off task, mind wandering, not like actively engaged. The default mode network has been kind of like consistently implicated in being having less of this kind of like within network connectivity, less of this functional integration. And this has been hypothesized to be one of the kind of key mechanisms of the therapeutic effect of psychedelics. In addition to looking at kind of like brain network activity on a high level, you can then also zoom in much more deeply and look at what's actually going on at the neuron by neuron level or the neuron connectivity level. Here, we see kind of very certain very consistent story around this psychoplastogenic effect, or this neuroplastogenic effect, where there are a number of different markers of increased plasticity in the brain. So one that's very common is this increased brain derived neurotropic factor, which we see hours after psychedelic use for a variety of different psychedelic drugs.

And this is then accompanied a few hours later by visible increases in the dendritic spine growth in cortical pyramidal neurons. So this is both the number of dendritic spines and the complexity of the spines. You can see a very clear example of this here on the right, where you have on the top the vehicle and then the bottom the DMT treated pyramidal neuron. And the difference is very, very stark. You don't even have to do any sort of actual computational analysis for it to be clear that there's quite significant dendritic growth from a single dosing. And the changes to functional connectivity, which I mentioned in the previous slide, tend to persist, though in a less extreme form than what we see during the acute effects.

So the question then becomes, given all of this, how can we attempt to tell a consistent computational story of what's actually going on during both the acute effects, the post-acute effects, effects, and in terms of the actual downstream therapeutic outcomes?

So what's kind of been proposed as the one of the leading models of this is the relaxed belief under psychedelics model, the rebus model of Carhartt, Paris, and Tristan.

And so this builds on the kind of like active inference hierarchical predictive processing framework where we can understand the brain as the series of predictors stacked kind of like hierarchically along brain networks. And predictions are being made by higher level networks that are attempting to predict the errors of lower level networks. And the process of basically doing this then produces this series of belief landscapes at different levels of spatiotemporal abstraction.

And these belief landscapes, of course, can encode beliefs with various levels of precision.

And it's the modulation of this, the precision of these beliefs, which is believed to take place under psychedelic use.

So here is kind of like a very high level version of this rebus model and how belief relaxation might actually happen.

Right. So given that you have this series of cortical networks, which are representing beliefs of different spatiotemporal scales, we can kind of think of the thalamus as being kind of like the entry point where sensory information is being gated into this hierarchical system. And the prefrontal cortex is being kind of like one of the highest level networks, right, which is then responsible for both representing the most abstract kinds of beliefs as well as being kind of the primary inhibiting force for all of the downstream beliefs in the network, in the system.

Right. So what is this heavy organism of the 5HT2A receptor supposed to be doing in this case?

Well, it's going to be decreasing the prefrontal cortex's ability to make coherent predictions and coherent representations of belief, which will then disrupt its ability to inhibit lower levels of the cortical hierarchy. At the same time, by disrupt the this activation of the thalamic or this disruption of the thalamic gating, right, is going to result in additional sensory information entering the cortex, the wooden underwise, into the cortex. You have these kind of disruptions of these two levels, which then result in beliefs being spontaneously represented in ways that they wouldn't normally be represented. And the rebus model, at least, is suggesting that this mainly is going to manifest itself as a kind of relaxation of these beliefs.

So here is a kind of like very simple diagram of what this might look like. So you have some landscape of beliefs here, and you can kind of interpret the position along the x and y, or I guess the x and z belief in a, as being corresponding to some particular belief that might be represented by the network at this layer, or at this level of representation.

And then the y-axis, the height, as being the actual precision of that belief, right?

So here we have this kind of very deep part on the right part of this network, around this landscape.

And this would be a very strongly represented belief, let's say.

What the acute effects of psychedelics are then understood to be doing, right, is to be flattening this landscape, to be less strongly representing these strongly represented beliefs, to maybe making it easier for the reversal of this landscape. So to move from one attractor state to another attractor state, which is meant to be understood to be kind of therapeutic in various ways.

And then the post-acute effect being some kind of return to normalcy, but one that is not totally complete, one that still incorporates some of these newly relaxed beliefs, these newly explored beliefs.

Building on this Rebus model, Karin Harris and colleagues last year proposed this canal model of psychopathology, which basically kind of takes the intuition that the relaxation, you know, if the Rebus model is correct, if the relaxation of beliefs has a therapeutic outcome, if you kind of take that way of thinking to its logical conclusion, you might then propose something like this, which would say, if relaxing overly strong beliefs is therapeutic, then having overly strong beliefs in the first place is pathological in some kind of way. And they refer to this development of these overly strongly represented beliefs as canalized beliefs, and this problem generally is canalization or this process of developing them as canalization. And this is the canal model of psychopathology. So they kind of build on this idea, this set of previous work in the literature suggesting that there may be some primary factor of psychopathology. And they suggest that this primary factor is canalization. And in this work, they discuss a number of different psychopathologies, which seem to quite clearly fit this canalization type characterization. So it's very relatively straightforward, I think, to understand things

like addiction, depression, anxiety disorders, obsessive compulsive disorders, all of these involve in one way or another. Having some strongly held beliefs, typically negative beliefs,
in the case of depression, anxiety, obsessive compulsive body image disorder, or in the case of substance use disorder, some very strong beliefs about what kind of behaviors should be taken at any given time, in any given context. And then of course, the effect of psychedelics is to undo canalization.
And the better able any sort of therapeutic intervention is to undo canalization, the more likely for a positive therapeutic outcome.
So I think this work has a lot of strengths, and there's a kind of elegance to it.
In our deep canals work that we have been, we've worked on, we kind of identify a couple of limitations of this canal model and attempt to extend and refine the model in ways that I think ultimately make it more powerful while keeping the essential elegant insight that canalization is intimately tied to psychopathology.
So the first limitation I think that we identify is that in the canal model, it's very heavily implied that all forms of canalization are not logical in one way or another. And as I mentioned just now, this is clearly the case
for a large class of very common psychopathologies. It's when we start to look more towards edge cases, I think,
that these exceptions start to problematize this understanding.
Right. So if we consider psychopathologies such as bipolar, borderline schizophrenia,
and depersonalization disorder, it becomes much more difficult to kind of easily classify these as simply problems of overly strong priors. In many cases here, we have problems of priors that are not so strong, or priors that seem to drift over time in ways that are problematic for both the individual and the people that the individual are close to. And if we think about psychedelics then as being this kind of panacea to undo canalization and undo psychopathology, we also run into this other issue, which is that there are certain kinds of psychopathologies which are typically
issues for use with psychedelics. So schizophrenia being a very clear one, where there's evidence of psychedelic use being a trigger for the onset of psychosis in individuals who are predisposed to schizophrenia. There's also evidence of the development of depersonalization disorders with strong psychedelic use. And this is not what we would expect if psychedelics were simply universally undoing or getting rid of psychopathologies. There's also this idea, which I think Adam is going to develop quite a bit later, that the stability of beliefs is typically correlated with positive mental health outcomes, not negative ones. And what is a stable belief if not one that is highly canalized in some kind of meaningful sense within the system.
And a second limitation is that if you think about an actual dynamical system, one that is parameterized in some kind of meaningful way, like a brain, which is parameterized by synaptic and then instantiates neural activity patterns over time, and we think about what kinds of optimization landscapes are actually being induced by such a system, we find that there's not just one kind of canalization. There's actually two kinds of canalization. And in our Deep Canals work, we attempt to take inspiration from deep learning literature to tease apart these two kinds of canalization and understand

them a bit more clearly. And they are, just to kind of give a preview, these two different things, overfitting and plasticity loss. So overfitting is the problem of you have, you're canalized in a particular kind of way, and whatever sort of policy you've learned, whatever sort of set of beliefs you have, these don't generalize to some other context. If you're in context A, you may be very well adapted, you move to context B, and whatever you've learned doesn't apply there.
Then plasticity loss is a little bit different. Plasticity loss has to do with one's, not one's ability to generalize, but rather one's ability to adapt or to learn. So imagine you're in context A, you may or may not be well adapted to context A. If you move to context B, the question becomes, are you able to adapt to context B or not? Not are you immediately already ready to deal with context B, but given time in context B, can you learn and change? And we use a very simple recurrent neural network model to kind of make sense of this. And I think this is important because there can be different kinds of canalization on these different kinds of optimization landscapes, and these have unique implications for the treatment of psychopathology. And I think even more importantly, psychedelics work on these different landscapes in different ways. And so understanding how they work is essential to developing better drugs, better treatment protocols, and more personalized treatment.
Okay, so what are these two optimization landscapes? How can we make sense of them?
This is kind of like the big overview figure of the deep canal's work. So we kind of start with this intuition of a recurrent neural network, right? And in a recurrent neural network, there are these kind of two sets of parameters that you have at any given time. You have the actual weights of the network, and then you have the hidden activation of the network, right? And so the weights of the network are theta, and the hidden activation is h, and there's some input into the network at any given time step. And then the output of the network is a function of both the input and the hidden state, and the hidden state is updated at every time, the hidden activation pattern. Great, and this gives us essentially these two different landscapes. One, what we refer to as the inference landscape or the type A landscape, and this is equivalent to the neural activation pattern of a brain at any time, right? So what gets read off from an fMRI, what gets read off from the EEG is going to correspond to this activity landscape.
But kind of like underlying or supporting this activity landscape is this other more fundamental landscape, which is the actual optimization landscape that is induced by the synaptic weights themselves. And this is what is the result of right-click synaptic connectivity in the brain between all the sets of neurons in the brain. This is what is changing when we have various forms of like heavy in plasticity, heavy in learning. And this is what kind of like instantiates the possible sets of neural activity patterns, right? So this is kind of like a way we can start to understand this.
Which is when we think about belief landscapes, right? As I mentioned before, you know, here's a very simple diagram where we have a two-dimensional belief landscape. And let's say there's, you know, someone who's depressed or potentially depressed and they might have a negative self-image, you know, I'm a terrible person. I can't do anything right. Something like this. This would be encoded as a particular position in this belief landscape. You know, when the neural activity pattern looks like this, this belief is what gets expressed, right?

And there's another, you know, set of neural activity patterns which would correspond to a positive self-image,
a positive self-belief, you know, I'm great. I'm wonderful. Or even just, I'm okay. I'm totally acceptable as a person. And this would be a different set of patterns.
So based on the synaptic, the actual connectivity structure of the brain, right, these two patterns may be maybe more or less likely. So here, these are being represented by these two different sets of neural activation landscapes, which are the result of two different positions in this synaptic weight landscape space. And in this top space, the one that's in the middle here, that then is represented here at the top of part C of the figure, we see that there's very high precision assigned to, or very high likelihood assigned to the negative self-image, and very low likelihood here assigned to the positive self-image.
But in some other part of the synaptic weight landscape, a very different kind of neural activity landscape might be instantiated. And this one kind of has the inverse properties, where negative self-image is unlikely, and positive self-image is much more likely. So we can kind of then understand positions in the neural activity landscape as kind of corresponding to specific phenomenological experiential instantiations of certain kinds of beliefs, right? So like the actual experience of holding a negative self-image or holding a positive self-image.
And we can understand positions in the synaptic weight landscape as being something more general.
So in the case of depression, there might be a specific set of synaptic weight connectivity, which would correspond to what you might call, you know, a depressed individual, someone who might qualify as having major depressive disorder, and another set of synaptic weights connectivity pattern that wouldn't qualify as that, right?
Okay, so before I get into the psychedelic, how psychedelics relate to this, I think Adam is going to talk a little bit about some work he's done extending the Rebus model.
So, yeah, I've been thinking about this a while, and then along the way, I met Arthur, who also has been thinking about this a while, but in different ways. And so, like Arthur's background is in basically consciousness science and a very peculiar field of computational consciousness science in terms of actual models of what it would be to have a conscious system, even potentially an artificially conscious system. And so, you know, I'm coming from more free energy principle and active inference in my background,
and Arthur's coming from more mainstream machine learning. And so we were coming together to explore this beast from different perspectives, and been finding a lot of, I think, very interesting overlaps.
And so, in terms of Albus, the basic idea is to take Rebus and expand on it.
I love Rebus. It's a great framework. And I've been trying to make it, I guess, more flexible in terms of being able to handle like a wider range of assumptions. So not just talking about relaxed beliefs, but also potentially strengthened beliefs and in different combinations. This was always part of the Rebus framework, and that you can have all sorts of both direct and indirect effects when you're dealing with a large predictive hierarchy or any, you know, complex system. You change something in one

way, but then you'll get an effect elsewhere. So you have this hierarchy of predictions. For instance,
you can potentially relax beliefs at higher levels, and that can indirectly cause strengthened beliefs at
lower levels. So you can both get relaxed and strengthened beliefs and psychedelics and all sorts of different combinations.

Dr. And so with Albus, the idea is, so I've been calling, as opposed to a belief relaxation, when you get
this sort of basically, bottom right here, you might see it. So this red would be basically
something like a synchronized complex of neural activity, and often involving the thalamus,
which would be thought to encode your predictions or your top-down stream of priors, which then are suppressing
the ascending stream of prediction errors. And so under Rebus, you're seeing here relaxed predictions, a smaller
complex of synchronized activity, and the gates are down, just like Arthur described, more things are coming in.

Rebus will sometimes be described in terms of...

Dr. So when we're talking about Rebus, this idea of relaxing beliefs, we're thinking at the
functional or computational level. So like David Maher, in analyzing the mind, you have these three,
I suppose, supervenient levels, aren't they? Compatible and synergistic. So computational, functional,
like abstractly, what is the system doing? Something like the pseudocode of the system,
and then implementational, mechanistically, how it's realized, and then in the middle, some sort of
algorithmic intermediate abstraction layer. How are these computations realized via these mechanisms?
Some bridging principle description. And so when it comes to Rebus, you can adopt it in different ways
and relate to it differently. So you can relate to it at the functional level, or you can relate to it at
the algorithmic or implementational levels in terms of where you're taking the account. So for instance,
you might think something like Rebus effects is a very valuable model of something like what they
call quantum change with psychedelics, but you don't necessarily believe in predictive processing,
but you still might be, you could say like a Rebus type theorist. Or you might be thinking in terms of
like Rebus, like in a more narrow sense of thinking of predictive processing information flows. But then
when you get to like the overall person and integrated organism, the agent, you might have different
interpretations there. So with the work with Albus, basically, the idea would be to take predictive
processing and models of psychedelic action and suggest that sometimes not only do we have these
relaxation of belief landscapes, but we might have the opposite of strengthening. And so specifically,
if you're activating these 5-HT2A neurons, which would be at the deeper levels of the cortical sheet,
and also the deeper levels of the cortical hierarchy more inwards and higher up. And these would be the
units that would loop to the thalamus and thereby be able to form large synchronous integrated complexes.
But then when you excite them too much, they don't wait for each other to sync up. They fire
whenever they can out of sync. And this relaxes the belief landscape. However, it's not clear this
should happen the whole way through for any amount of increasing the gain on these neurons. So in theory,
just increasing the gain a little bit, unless that might help them to sync up. That might strengthen

the belief, whether it's a situationally specific suggestion of you in this context, or maybe something old coming to the surface and, if you will, manifesting. So Albus basically, in a nutshell, would be saying to take Rebus and keep going with it and look for different combinations of both strengthened, both directly and indirectly, strengthened and relaxed beliefs. That's basically it. And so I'm not going to go into too many more details. Arthur and I just submitted this for publication. Wish us luck. Hopefully the reviewers will be useful. Maybe even kind, but preferably useful. And yeah, I'll kick it back to Arthur. But before I do, like this work, I think it's immensely important that we get these details right. These are extremely powerful medicines that could revolutionize things, or we could have a backlash if we do it wrong, if we're careless, if we don't see the risks.

Arthur and I think it's really important to be able to do it wrong. But also, if we're overly beholden to like a single way of viewing things prematurely, we might also miss opportunities to help, potentially enormous opportunities. For instance, like you might not think of helping people, like for instance, microdosing for cognitive decline. You might not have thought of that. And that might be a really big deal that you do think of that.

Arthur Worsley- So I guess before going back to Arthur, one more thing would be just in terms of, at least for me, like experientially, like the way I sometimes think of it is like with relation to wellbeing and suffering, both things matter. Like the strength of your positive beliefs and the strength of your negative beliefs. And sometimes the negative beliefs have positive aspects and vice versa. So it's both like, you might relax your beliefs and like, to use like a Harry Potter metaphor, like be somewhat like out of the, so much out of the grip of the Dementors that usually, that under some circumstances might be pursuing you, that you have room to be different, to become different, to think different things under this different state. But additionally, and not mutually exclusively, not necessarily incompatibly, maybe synergistically, the strength of what beliefs you're able to have in that moment, and maybe draw upon from the past could be deeply important. Your values, the kinds of beliefs that you call upon in times of need, and they give you hope. Maybe those are just originally suggested to you right there in the session. But maybe there's something else, another belief in that strengthening it could be a good thing. And even without a clinical context, like microdosing, for instance, it's not clear that exactly they're trying to relax their beliefs, like the range of use cases, the way we want to relate them, how we want to be in the world. It seems complicated to me. So I will get back to Arthur.

Arthur Worsley, Ph.D.: Great. Yeah, thanks so much, Adam. Okay, so with that context of Alice, I want to talk a little bit about how we might understand at a very high level, what psychedelics are doing to these two belief landscapes. So before I showed you very quickly, I guess I can go back to it. In the Rebus model, you know, there's only one type of optimization landscape. And so there's only one type of canalization. And what is supposed to be happening is a relaxation of beliefs, which is strong during acute effect and persists in the post-acute effect.

Arthur Worsley, Ph.D.: So with Albus, though, we might be able to think about all of this a little bit differently.

We can, in during the acute effects of psychedelics, interpret what's happening to the type A landscape as being less of a kind of universal relaxation and more of a kind of destabilization, where under certain circumstances, beliefs are being strengthened, other beliefs are being relaxed. This is changing quite rapidly over the course of time. So, you know, a certain belief might be transiently strengthened.
A few minutes later, it may no longer be supported or supportable. And this actually functionally, you know, enables a kind of exploration of all the of a wide space of belief instantiations during the acute effects of the drug. As far as the inference landscape is concerned, you know, post-acute effects, as Adam was saying, it could be the fact that certain positive beliefs are explored and can be strengthened. Other negative beliefs might be weakened. This is kind of like the ideal psychotherapeutic outcome. It may be, you know, on average, that something like a relaxation, a general relaxation of beliefs is what you might see. But the way that you get there is likely going to be through a kind of much more complicated process. I think what's then interesting and related also to what Adam said about thinking much more broadly about how psychedelics might be used in terms of neurocognitive disorders, in addition to kind of like traditional psychopathologies, is to think about what's happening to the learning landscape, the kind of like underlying synaptic way landscape. And what do we understand is happening here kind of quite clearly is there is neurogenesis, psychoplastogenic effects, there's dendritic growth. And what is this enabling? Well, this is enabling greater connectivity between within and between networks. And so kind of from a functional learning perspective, learning theory perspective, what this means is that gradient information can flow more easily through these networks. And what this means is that it's easier to get from one part of the network to the other part of the network, there are fewer of these kind of optimization, local minima, there are fewer of these kind of like tricky saddle points. And what this means kind of like functionally is that this plasticity loss that I discussed earlier is going to be less of an issue.
And here it seems to be kind of like regardless of the acute, the actual phenomenological effects, you know, there are very strong downstream effects on this structural connectivity. And I think, you know, there's some interesting work suggesting that there may actually be independent mechanisms that are affecting these two landscapes. So there's recent work suggesting, for example, that psychedelics as well as ketamine is able to kind of like particularly bind to receptors in the brain that produce, that are results in the kind of release of the brain derived neurotropic factor, and then the downstream changes in canaptic weighting, and that this may be potentially totally independent of the 5-HT2A agonism, right? And this has very interesting implications for future drug design and development. Okay, so I'm going to, before my next slide, turn it over to Adam once more to briefly talk a bit about personality theory and the relationship between psychedelics and personality. Hello again. So, um, so for the past few years, I've been doing some work with a personality researcher, Colin de Young, on intersections between the way agency is modeled in the free energy principle and active inference framework, and the way that persons are considered in different situations in personality science and looking for points of intersection. And so, um, you know, it's personality, personality is actually an extremely deep concept. And if you think of it, it's, um,

I guess we call it like the normal form of a system in terms of it's, uh, what is the way you can describe the most of the system, um, with like the minimal message length, the, um, account for the most of the least. And what we're trying to account for is the states that people tend to occupy. People aren't always happy. People aren't always sad. People aren't always, you know, friendly. They're not always angry. It's all sorts of things that are there. There's a range that people are encountering, but there's, they tend to occupy some states more than others. And you might call that the person, their personality. And you might call it the person, many systems would have their own personalities in terms of trying to describe something like their essence. And so, uh, they're attracting states. And so with, uh, Colin, the young's work, um, he interprets personality and these, um, major domains of personality that people have discovered, uh, uh, a reliable factor structure ways you can chunk the variance, uh, the big five as a trait hierarchy, but where these different traits are interpreted as cybernetic control parameters, uh, specifically, um, cybernetics in terms of the study of goal seeking systems governed by various forms of feedback that different stages of, uh, the process of pursuing goals, uh, evaluating outcomes, updating, um, that, and you understanding that basically the personality as different modes of being in the world for cybernetics systems, interpreting them as parameters for goal pursuit, because we are goal seeking systems.
And so I guess moving to this, the, and with relation to what we're talking about now, both in terms of personality formation, reformation, and the structure of personality, um, the P so, okay, to get to the P factor, uh, within personality theory, this big five traits, the set of big five traits, there's a, uh, richer or more detailed, uh, factor structure, um, a hierarchical, uh, structure for personality that's been discovered. Um, so above the big five, neuroticism, agreeableness, conscientiousness, you will have a meta trait of their shared various that's observed called stability. It's also been called alpha by, I think, Digman. Um, and then above, uh, the traits of extroversion and openness intellect, you have this meta trait of plasticity and, um, or called beta, uh, and the shared variants of these two. So what's interesting is that within, uh, the, the field of, um, within high top, which is this effort to try to characterize psychopathology using the same kind of, uh, data-driven methods and personality science, a principled account of the structure of psychopathology, the way that personality has a structure and trying to account for this, there seems to be a common factor across, um, many forms of psychopathology that's referred to as the P factor. And it tends to be the inverse of stability. It's high neuroticism, uh, low agreeableness, and low conscientiousness. And this is actually, uh, uh, what's referred to in the article, um, as, um, this P factor, it, thinking of it as a lack of flexibility, um, you could make a case for it, but more straightforwardly, it seems to be a lack of stability. For something to be a lack of flexibility, you would maybe be thinking of something more like, uh, plasticity, like someone having low plasticity, but empirically, that's not what's observed. And so if empirically, that's not what's observed, uh, that's a very important detail we should stop and take note of. So you might say that, um, rather than emphasizing the first principle component of psychopathology, it's possible that this canal model is, rec, is actually focusing on the second factor. Um, it's not, you know, it's not a contest.

Uh, it's still enormously a big factor, like the work of Stephen Hayes and others showing that, like a lack of flexibility is transdiagnostic, but just empirically speaking, the first principle component of psychopathology might be more closely related to a lack of stability, um, rather than insufficient, um, plasticity. Um, well, I guess one more thing to think of in terms of stability and plasticity, like, as I said, it's not like a, um, it's not exactly a contest. Uh, they're, um, they're both important, both need to be considered and they're deeply interrelated. So Colin describes them as being in a state of something like dynamic tension. Like if you're insufficiently, uh, plastic and then flexible, you're not going to stay stable for long in a changing world. Uh, but if you don't have a stable base to move from, uh, it's going to be hard to be plastic and reconfigure yourself in a sustainable way either. However, you could also see someone being over, um, overly rigid and excessive stability in a way that undermines their plasticity or overly flexible in a way that basically is destabilizing. So a dynamic tension, but involving synergies also. And, uh, I think in terms of the structure of personality and wellbeing, uh, this is what we need to keep in mind that both are needed stability and plasticity. And this is, uh, um, also just within, um, it's, it's, it's actually what this was inspired I think of in Collins thinking to consider these meta traits this way, uh, from machine learning. I think actually reading Grossberg and stability plasticity trade-offs, and that's why he interpreted these meta traits in those terms. Uh, so I guess one more thing is, uh, um, there is a Petersonian backstory here. They don't know if I want to get into, but we can discuss it later if there's interest. So yeah, back to Arthur. That was great. Thanks Adam.
Yeah. Very, very helpful context, which should hopefully make, uh, this slide here much more clear to everyone. Um, so given these kind of like two meta traits, uh, stability and plasticity, which as Adam, you know, beautifully described here in this kind of like dynamic tension with one another, um, we can kind of go back to these two optimization landscapes and these machine learning concepts that I discussed earlier and try to understand how these things might all relate, um, and ultimately, you know, relate to psychopathology. So I had hinted before that there, um, were these kind of two types of canalization we might say, one being overfitting and another being plasticity loss. And we can interpret these as kind of like being extremes, uh, the kind of like canalized end of, uh, how these two optimization landscapes might manifest. So in the type A inference landscape, this looks like overfitting on one end, heavy canalization and underfitting on the other end in the machine learning context, right? So overfitting being, uh, having very strong rigid beliefs that are in a particular environmental context, adaptive, but are very unlikely to be adapted in any other context. Underfitting being having a set of beliefs which are, uh, not adapted to the current context and also unlikely to be adapted to any other context. Awesome. As far as stability is concerned, I, uh, I think we can understand stability. This meta trait is kind of representing the balance of these two things, right? You have high stability when you are neither, um, overfit to a particular context or underfit. Um, so this means you're well fit to this context that you're in now, and you're also likely to be well fit to other future contexts. Um, I think this is also in complexity science sometimes described as meta stability, right? Um, so you're in a stable point that'll also likely be able to

take you to other stable points in the future. And this kind of like ability to stay stable over time is then, uh, stability meta trait. We, uh, can then look at the type B, the learning landscape, where the kind of extreme version of canalization is this plasticity loss where it's very difficult to adapt to new contexts. Um, the inverse of this is then catastrophic forgetting, where the problem is not one of adapting to new contexts. The problem is one of meaningfully retaining adaptive strategies and beliefs from previous contexts. Um, and so obviously, you know, neither of these are desirable. Um, what is desirable is some kind of balance between these two, right? Or one is able to adapt just enough to new contexts without being, uh, without forgetting previously learned things. Um, but here, the relationship, uh, to the meta trait of plasticity is a bit different than stability, right? So as it is being represented on this graph, um, you know, the more plastic you are, basically the less canalized you are in the space, the more easy it is to move around. Um, but actually extreme versions of this end up manifesting as catastrophic forgetting, at least in the machine learning context. Um, and I think this in particular has relevance for, um, psychopathology and relevance for this idea, maybe complicates this idea, right? That, uh, psychedelic therapy is, uh, in and of itself likely to always lead to positive outcomes. Okay. And we can attempt to understand this by, uh, making an effort at trying to map or understand, um, various forms of psychopathology and the context of these two different learning landscapes. Um, and so before I get into this too much, I think it's important to say that, um, this is very much a kind of like preliminary and provisional attempt at making sense of these things. Um, I hope that this is kind of the start of a, um, a much more fleshed out dialogue, looking at how we can kind of understand psychopathology with respect to machine learning concepts. Um, but I think this at least gives like a good starting point for maybe trying to understand where and why, um, psychedelic therapy is useful, isn't useful. Um, which is a bit more nuanced than this simple idea of like psychopathology means canalization and psychedelic therapy means reducing canalization, right? Okay. So let's, uh, try to make sense of this a little bit. So for the kind of classic set of psychopathologies for which there has been a lot of great research done with psychedelic therapy and a lot of clear benefit has been seen. Um, I think you, we can kind of class these within this upper left-hand corner here where these are disorders of both overfitting, which means in the moment, overly strong beliefs get represented. You know, these are strong attractor points in the neural activation landscape, um, and, uh, plasticity loss in the type B landscape, which means that not only is a certain kind of dynamics instantiated, right? Which may, for example, in the case of major depressive disorder, extremely negative self-evaluations are uncommon, but it's the case that even given the correct changes in environmental context, even in the, you know, even given, uh, in the case of, you know, psychedelic system therapy, a lot of the work, um, with major depressive disorder is being done with treatment resistant to depression, right? And what is treatment resistant depression, but, uh, a kind of a set of beliefs, a set of neural activity patterns, which are um, resistant to adaptation to new environmental context being, you know, whatever sort of psychotherapy or previous drug was used. So this is high plasticity loss, high overfitting, right?
I think we can then, uh, and these are, you know, kind of fortunately, you know, for psychedelic therapy,

and I think maybe for psychiatry in general, these are all very, very common and very, very costly, um, psychopathologies, right? So if it is the case that psychedelic therapy can treat these effectively, that's fantastic. Um, but I think we can maybe look at a more complicated, you know, if we start to expand other psychopathologies, um, the story becomes a little bit more complicated, right? Um, so let's look, we can look at the bottom left-hand corner here where we have cases of, in the inference landscape, still overfitting, but, um, in the learning landscape, something else happening, uh, form of kind of catastrophic forgetting or form of kind of drift of the, um, the belief landscape that's being instantiated here, right? So I think two kind of representative, um, pathologies here kind of typically referred to as some form of bipolar disorder or some form of our borderline personality disorder, right? So in both of these cases, in the moment-to-moment, uh, neural activity as it unfolds, there are very strong sets of beliefs being instantiated at any given time. And these are very difficult to change, right? Um, but the key differentiating point here being that over longer timescales, over the timescales of days or weeks, um, the kinds of beliefs that are instantiated change quite a lot, right? So if someone who's bipolar, if they're in a manic episode or in a depressive episode, the nature of the belief landscape being instantiated is very different, um, for borderline individuals from a daily or weekly basis, the nature of one's beliefs about the people that are close to them can change very rapidly, um, very extremely, very, in certain cases. And so here the benefit of psychedelic therapy, uh, is a bit more, uh, mixed where there is likely the case that to the extent to which the, uh, kind of overt type A beliefs can be updated and changed to be more positive, be hopefully more relaxed. Um, that's positive, but to the extent to which plasticity may be being increased when it's not actually needed, when there's not actually like a meaningful therapeutic benefits of that. Um, it's a bit questionable. I think, you know, there's clear, um, borderline or rather bipolar, for example, is often, uh, individuals with bipolar are often excluded from psychedelic files, um, under the assumption, for example, that psychedelics may trigger a manic episode or may trigger psychosis. Um, and I think that is, uh, a valid concern. Um, so on kind of that topic, we can, then this bottom right-hand corner would be cases of both underfooting and catastrophic forgetting, with, I think, schizophrenia being kind of like the most clear manifestation of this. Um, and here, um, you know, there's very little work currently looking at trying to treat, uh, schizophrenia with, uh, psychedelic therapy, I think for pretty good reasons. Um, though I would say, um, to the extent to which there are secondary, uh, manifestations, uh, secondary issues like depression, anxiety that people with schizophrenia might have, you know, these actually may be treatable, um, with some form of psychedelic therapy. And then lastly, this, uh, upper right-hand corner here, uh, would correspond to underfitting in the type A landscape along with plasticity loss. So I think a good example of this is something like ADHD, um, where there are not very, uh, particularly strong beliefs that are being stantiated at any given time. Um, but it's difficult to change these set of beliefs nonetheless. Um, I think here there's a case, uh, where it's not totally clear, um, how the benefit will actually play out. Um, in both of these cases, uh, you know, I've heard interesting work around like microdosing may be of issue, uh, interest. Um, I think, yeah, more research has to be done to

be seen, right? And then what would microdosing be doing, right? Microdosing would be like very slowly, uh, kind of like decreasing plasticity loss just a little bit, um, potentially also, um, decreasing underfitting, right? As Adam had kind of suggested in the ALPUS model.
And here we have just kind of like a brief overview of more or less everything, uh, I've described about how these two landscapes are different. Um, I think most of these things I've, I've said before pretty clearly, right? So the type A landscape, um, it corresponds to the neural activity pattern.
This is the hidden state of the RNN. Overcanalization is going to be stereotyped mental circuits, very strongly held beliefs. Undercanalization is going to be these inconsistently deployed circuits. Um, it looks like, uh, overfitting in the canalized case, underfitting in the non-canalized case, and then correspond to stability, the meta-trait. Um, I hadn't discussed this before, but, um, I think there's a relationship between cognitive flexibility and this inference landscape.
Um, psychedelics are mainly going to affect the, uh, work at the acute level care.
And this is by directly exciting the neurons or inducing forms of hadibian plasticity.
And the outcome is going to be a mixture of relaxation and strengthening of the kind that Adam described in the, the ALPUS case. The type B landscape then is working on the synaptic weights.
And these are the actual network parameters or RNN model. Um, overcanalization is this inability to learn or adapt over time. Undercanalization is the inability to retain information, right? And those correspond to plasticity loss and catastrophic forgetting. This also is kind of the traditional plasticity meta-traite, um, which I think corresponds to this more, uh, this longer term, uh, constructed psychological flexibility. Um, the effects are going to mainly be mediated post-acutely with psychedelics.
And these are the more downstream meta-plastic effects, um, the psychoplastogenic effects.
Um, and I think on the whole, the outcome, the effect here is a relaxation of canalization.
And, uh, yeah, these are just the references. So those, those are the slides I have. That's kind of the overview of the deep canals model. Um, I think now I'll turn it over to Mike, um, who's gonna share some thoughts both about this work and some of his own work and kind of this, uh, approach to understanding psychedelics more broadly. Uh, great. Uh, yeah. Thank you for the, that overview. That was great. Um, so I guess, uh, I'm just in general, very, very optimistic about this corner of, uh, of neuroscience that, uh, there are these very intuitive models, uh, that are very sort of clinically significant. And, uh, and I also think, you know, there, there is a lot of responsibility to sort of get things right here, uh, as well as, as Adam noted. So, uh, yeah, I, I have just a few comments, um, and, uh, for the, for the viewers, uh, I, I'm not speaking, uh, for, for Arthur and Adam, um, I'm just grateful for, uh, the invitation to join the discussion. Uh, so I wanted to, to note, uh, quickly three research themes dealing with, uh, the implementation of this class of model. Uh, so just broadly kind of pointing at, uh, both deep canals and, uh, and rebus, uh, and, uh, and albus. Um, uh, so, um, in 2019, I wrote neural annealing and that was sort of, uh, developing the canalization frame within the context of energy buildup in the eigenmodes of a system. Um, so you have the, the nice annealing, excuse me, annealing metaphor. Uh, you have a temperature parameter and entropic disintegration. Uh, uh, uh, Robin Carrotera said, uh, uh, a beautiful paper on that, um, uh,

tentropic brain. And, uh, then you, you sort of have a, a post, uh, heating result, uh, where the microstructure of your system, uh, is sort of a, a result of the, uh, sort of how hot and how long the, the heating was so it's it's an alternative take on belief relaxation
and i do think that like there are some pluses in terms of sort of explicitly making the annealing metaphor um and uh i think that's uh like two things that i'd point out uh one top-down models or top-down predictive models as energy sinks um and eigenmodes as the places where energy can build up for almost sort of hide from top-down models um so you get this uh this phenomenon of semantically neutral energy or semantically illegible energy you can say uh which is sort of kind of how uh perhaps uh psychedelics music and rotation all sort of allow energy build um the the energy that they contribute to the system can evade the top-down models um and uh yeah just to note one item that i'm looking at now is what is um deep canals has the inference landscape versus learning landscape distinction and how would one sort of build that distinction within the neural annealing framework and uh i guess one one metaphor that i'm i'm considering is the self as sort of a magnet uh to sort of uh apply uh a constraint a uniform focusing constraint on the system as it uh is cooling um the second uh item that i i wanted to mention uh so i i have this piece autism as a disorder of dimensionality and uh this brings in the concept of network density uh that it it may very well be that uh people on the spectrum have literally more neurons and or more synapses and there's this great michelangelo michelangelo quote uh i saw the angel in the stone and i set him free um and sort of the thicker networks you have the more stone you have to work with but you actually have to do the work um so uh you you sort of have to have to do sort of uh well normally you sort of inherit these highly pre-trained optimized circuits almost like asics application specific integrated circuit um and then uh this is kind of more uh an untrained network and you have to turn it yourself um and uh within the deep canals framework this would show up as both under fitting and higher capacity for plasticity uh so i think that's uh that's going to be a an interesting thread to pull um finally uh so i just posted that last week principles of vaso computation and as an implementation level account of active inference uh and this is a a big rabbit hole um i i don't think we have time to to go down today uh but i just comment that i really like the way deep canals explicitly splits out the learning landscape and the inference landscape that uh i think that making this distinction is increasingly going to make sense uh back to back to arthur yeah awesome well thank you all for the presentations arthur and adam feel free to to pick up where michael left off or i can bring in some questions from the chat or from what i've written down all right just uh yeah just a quick comment um i do think that uh in in the deep canals framework uh it's going to be interesting to sort of look into uh metabolic demands uh and like sort of different uh like
metabolic disorders are are pretty hot in brain science right now and i do think that uh it'll be interesting to to look at sort of what the metabolic profile is in terms of each of the four items in a quadrant if one quadrant or another quadrant is particularly associated with uh
more metabolic demands in the system
uh

are there yeah
yeah no that's fascinating i was i'm actually i'd love to kind of hear a little bit more about this
comment you made about this as a magnet that's applying these constraints
oh yeah so there's uh
uh the like psychedelics the the effects are not or sort of the you sort of get pulled back into
position after after some time and uh
uh you know i i can't say that this is a full full theory uh at this point but um in the neuroscience
meditation 2018 i talked about um the self as kind of the the god of the gaps in terms of aligning different
scales of uh like you you have circuits which kind of pull in different ways you have the mesoscale
systems which pull in different ways and you have the the overall brain wide systems that pull in
different ways and kind of the self is sort of the thing that all tries to make everything work together
um and an alternate frame would be sort of the self is just kind of this
uh slow gravity toward disalignment
uh yeah um i don't have uh anything particularly uh more to share
i see yeah that looks like it's interesting
well to combine the heating up and the magnetism different metals and connections there to alchemy and
everything different metals when heated up and cooled down in specific ways changed their
magnetic profile so of course this is all just concordances but they're very interesting and
then also to the um the metabolic angle that reminded me of the um the trilogy by dale pendell
with uh pharmacognosis and and the other two books were treated alongside traditional psychoactives we
could say like caffeine and nicotine and some of the other substances that were mentioned here also there's
tea sugar and these are treated as psychedelic or at least as a in the same pantheon of neurocognitive
modifying in the sense that the um destinies of individuals and of of empires ride on the uh sense
making and decision making influence of sugar for example so it definitely broadens our perspective
to include the the so-called classical molecules like lsd but also maybe we could think more broadly about
um antioxidants or about different lipids or about different metabolic um connections
um michael pollan's uh recent book uh this is your mind on plants i think like it kind of has that
what you're saying in mind because like it includes caffeine um not like introduced as a psychedelic but
just discussed among uh also psychedelics i think our memory focuses on masculine but um just it is a
different way of being in the world uh potentially radically different like if caffeine is in your life or it is
not um it's a different life world
and different extended niche um i'll go to a question in the live chat so anyone can can give a thought
so matt asks is there any way how to approach the individual reactions to psychedelics in terms of positive
or negative results of treatment using the ideas of deep canals so how can we approach predictions and
explanations at the individual scale with what you're sharing today
well just one thing for wow
this is kind of like a meta comment but like if we're actually going to do um computational psychiatry
and we actually are going to and precision medicine so the the goal being to create models of sufficient

detail that they could actually guide our practice and guide our practice in a way where we know for an individual what they might need in any given context so precision medicine with respect to like standard antidepressants um you know people sometimes they'll go for a long time before they find ones that work and sometimes things are made worse and they might deal with antidepressant withdrawal as they're looking for the right one and if they're in a state of acute distress that could be um a hard lot and so like if you can help to speed up that process of finding what works uh the ability and what might hurt also to avoid that i mean the ability to help people is immense and so for that it's um like part of this so the meta comment is like um i have to just say i'm not convinced that uh an active inference uh we're doing the job i i think we're not acknowledging our uncertainty and i think the models are under specified with respect
to the um complexity of the effects we would expect on different scales and in characterizing the systems and we need to be more patient and modest but what we don't know before we've gone rushing in headlong into telling the rest of psychiatry how they should comport themselves and what models they should use uh based on psychedelic science and plausible models hopeful models but very early days and with the reality being complex in the ways we're seeing here like this was not an easy set of lectures you know or a set of talks like what we were going through this is like and yeah if we're actually going to be able to handle these powerful medicines responsibly we got to go into the weeds and so i did so the one more meta comment was like i think i saw like someone say oh how can something like help everything you're like yeah it's almost i don't know if this is right it's almost like a violation of like no free lunch theorems or something it's like how could something just be good across the board without you know something is off you've calculated wrong somewhere um you had to pay the price either in like some trade-off that wasn't acknowledged or but that being said there are some things which are generally good for a lot of things like being more flexible or like if if you can get something like meta stability like like i said there's almost like you know i'm i'm i think i'm a church member of the the i'm a faithful member of the church of criticality i think mark miller might have uh indoctrinated me with like optimal grips but basically like the ability to have like an optimal grip on existence in your soul not too tight not too loose and being able to move around and be like flexible and itinerant it's like something kind of like mindfulness is just like sriracha for the soul it's good for everything almost you just you know put on this put on that makes it better i i i i i buy that there might be some things that are good across the board but i wouldn't call though psychedelics necessarily um putting you in this like state of optimal criticality or anything it's like in general like the rhetoric i'm uncomfortable with of saying psychedelics do x i would prefer us to say like um which psychedelic psychedelics tend to do x but just a little more qualifier like not weasel words but just like a little bit more like uh uh this psychedelic might do x in this context with this dose and this sudden setting but that being said there do also those seem to be across the board um broads um sweeping not super but overarching statements that we can make that express a lot with
a little and so like i don't want to be so much of like a splitter and like uh a prickly person finding

all the details that like these powerful things we can basically give to people to guide like our
research i got i don't want to leave that on table either so yeah so i guess um uh that's a bunch of
metal level commentary that did not address the thing so i think that's useful it's very helpful i think to
have all that context um with respect to the question itself i would um maybe offer this which
might be helpful which is you know i think you mentioned it adam and it's very frequently mentioned
this idea of sudden setting right as being like heavily predictive of the quality of the acute
psychedelic experience and often also to some extent the quality of the post-acute experience um and i
think we can understand this in the deep canals model quite quite straightforwardly um so i'll just
quickly pull back up one of these slides here um yeah right so what is set and setting so um set is
mindset which would basically be some combination of the currents uh your current position in the synaptic
white landscape and the current state of your neural activation landscape right so your place
in both landscapes is your mindset and uh the setting is the what is generating this input right what
is generating the x into the network and that's you know some complicated function of the environment
right um and then we can understand right the evolution of movement through the inference landscape
during the acute effects of the drug as then kind of like very directly and straightforwardly being a
function of these two things right um because both of these these things are are basically what's
affecting how this landscape evolves over time and how you move through the landscape um and so you know
with this in mind i guess we can then start to characterize and think about like okay well what kinds of uh
you know we can maybe be a little bit more mechanistic about like how does input from the environment the
setting actually given you know that we relatively fix the actual synaptic way landscape how is that
changing the dynamics over time or the other way around right like how are uh is fixing the input
the environmental context relatively stable um and looking at individuals who are at different fixed
points at different points of time um going to change the dynamics of the event right and i think
um given that this is a highly stochastic process given that these dynamics of albus are are at play right
where like certain beliefs are being strengthened other beliefs are being relaxed um i think at least
having this hold on okay well sentence setting is really important and it's important because like
these are explicitly conditioning the way in which uh both the landscape the inference landscape is shaped
and how you move around it and how it's being you know dynamically perturbed um yeah hopefully that
answers a little bit the question nice uh i just want to jump in there just thinking about um biomarkers
that uh i mean if if we're looking for something that's of high clinical specificity uh we want to sort of be able to measure what's
what's going on uh in a very careful in a very careful way so like first of all what is temperature parameter uh you know we we
talk about okay psychedelics increase the temperature parameters but um like how high and do do different
substances and dosages uh sort of differ in interesting ways probably yes uh and then also sort of measuring
temperature parameter measuring the temperature parameter in different parts of the body so you might
have a different uh have a different uh temperature uh like not literal temperature but uh kind of

computational

like entropic concentration uh in different parts of your body um and also um the plasticity status in different networks throughout your body so i guess uh i see a big frontier here uh as thinking some doing some careful thinking around um biomarkers and proxies and inferring these these sort of things that we we are talking about as core and i think they are core and we should be able to find good uh good marks for them

yeah that's totally correct and yeah i just want to like repeat it for my own occasion right which is

that like yeah these two things which i think were maybe one was is i think already a core part of the

deep penouts work right which is that i was showing these single images of these toy optimization landscapes

but of course um the human being the nervous system right uh has multiple many of these landscapes

hierarchically organized right um and they influence each other in a complicated way um and the kind of like

yeah the project for it is to understand where these landscapes are what they look like and how we can

find biomarkers for them yeah and begin you know if these two kinds of characterizations are true

um this overfitting underfitting plasticity loss and catastrophic forgetting then it would it's a very

meaningful project to be able to you know quantify that uh objectively somehow an individual and that's when

you can start to get to this point that adam was talking about of like precision medicine and

understanding like okay we'll actually attend that right like these networks are catalyzed in this

kind of way and because of that we can say with much more certainty that this kind of you know

psychedelic intervention is going to likely be helpful so i'm wondering i don't know if this like

potentially brings in some of the work with like um uh vascular and neuroviscilar uh neurovascular and

visceral integration that you're doing but like so there's like a sense in like the franergy principle

where the overall phenotype and an extended phenotype in terms of its cup it's constructing and coupling

within constructing its eco niche there's um its success and manifesting this phenotype um it's there is one

for energy functional for this overall effort as a kind of prediction that's enacted through the

dynamics but you could also like break it down in terms of like subsystems with like local gradients

and like local autopoetic processes that are in like they're brought together into an overall system

hopefully with synergies um but and they have to be to be an organism and an agent we call this like

a thing uh to the extent it has thingness it's coming together but i guess so you could i guess

think of maybe like uh energy landscape you know something like uh let's just say like right now in

this moment like my uh explicitly felt beliefs but my my body also though might have a different opinion

different parts of it and it in terms of like what it's implicitly optimizing it's revealed preference from

it's an action that may or may not correspond to how i feel about it i might not it might be difference

of opinion i might not even be tuning in i might be ignoring it because i don't want to and so it's

kind of like the score kept by the body might have its own um uh be describable as its own like

information geometry which is being tracked two degrees and is in different degrees of uh alignment with

um different ones at different levels of a control hierarchy i have some plausible hey absolutely uh

i i think so um yeah uh and like i i think of i think of the body as sort of um uh a parliament of

different uh networks and uh you know alignment is the scarce resource coordination is always a scarce

resource resource and uh like how does the body keep alignment and when does it go wrong too and i guess
i i see this as sort of um i mean harmonically mediated each part of the body has some choice of
what else in the body to listen to and depending on the particular sort of location in the information
topology uh different parts of the body have different carrots and sticks by which to try to control
other parts of the body uh so if if your stomach is very unhappy uh it's gonna start grumbling and
then it's uh it'll it'll let the body know okay i need some food like i really seriously need some food
um and then the rest of the body they can sort of turn away from it uh and then it'll just get louder
uh and then you have to turn it away from it more and you'll get louder and that's a of course a
negative dynamic if it gets what it wants maybe it sort of gives a very harmonious hum uh and uh and
sort of tries to give give carrots the rest of the body so yeah i just uh there's an ecosystem there
i'll go to another question in the live chat so bert asks
what do you think of the average of solving within a generative hierarchy and the impact of temporary
openness moving it upwards and then i asked to clarify what do you mean the average of solving
and bert wrote where predictions meet incoming signals like discussed in rebus where top-down
predictions are softened in precision and thus and thus solved for
oh
i think arthur will be able to get into the details of that um
probably better than i would but the uh one thing i'm wondering there is um
sometimes like like like again i love the entropic brain uh but sometimes i wonder in terms of like
the connections whether i don't so how we're having more or less entropy and whether that matters the how
so for instance like you can think of like a free energy landscape that's been flattened or you can
think of one where like you're better able and you have like more like energy for like this uh you know
the manifold activity is better of like climbing up or down the the peaks that are there for the more
enduring belief landscape so it's like in the moment it could be that um uh my previous way of thinking is
unavailable to me and i no longer believe those things at any level um or i'm i i can uh support
more novel inferences of that i wouldn't otherwise i can kind of move into an area of of mind space that
would otherwise be difficult potentially because of a of greater energy into the system greater throughput
noise
yeah so as i as i think i understand the question right it's basically asking about
trying to optimize for this free energy um
variable and in the con the context of deep canals right like this is the loss function
that's driving the optimization of the learning landscape the synaptic way landscape right but
like the entire connectivity structure of the nervous system is like trying to get to this point
of like low free energy and i think there's lots of uh there's you know for any given individual in a
given environment assuming the environment is relatively fixed right there's like an optimal
um there's a global optimal and it's very likely that the individual is in some local optima um that
is maybe close to it maybe far from it um but that's just inherently you know um what we're you know
within the context of active entrance that's what we're assuming that the nervous system is doing

is is doing this gradient to some in the space and openness then is going to uh course
respond to this plasticity meta trait um or be one of the components of it and it is in the deep
canals model then going to correspond to an ability to move around this optimization landscape more
easily right it's going to be a flattening of the landscape or more precisely i guess a smooth
in of the landscape um such that it's less likely that the individual the nervous system will get stuck in
these local optima so maybe it won't mean that the individual will arrive at the global optima i think
it's difficult to say what that even you know might be or however our individuals might get to such a
place at all but i think it is likely that the individual will be will be more likely to find a more
uh a more global optimum and more easily move from one uh option to another
uh and i think that's kind of what this this openness is corresponding to um
yeah okay uh oh i so in in neural kneeling i uh i mentioned the uh the idea of psychedelic
extrapolated volition kind of a play on a coherent extrapolated position from uh from the ai safety
crowd and it's this sort of sense that um if temporarily you have more control over sort of
low level shaping of of your your cognition your emotion um and you know maybe sort of based on this
this attractor that we call the self uh or you know i i guess like based on what is a good question
um then what are good signs that you should be trusted with that power or you should trust yourself
with that power rather and what are what are telltale signs that you shouldn't trust yourself with that
power and um no i i don't have an answer but i think it's an interesting question yes i am normal
and can be trusted with the key to the doors of perception let me in um i have a short question
where is action in this model
so um
imagine many places arthur get into that but for um what i was thinking with um albus is that actually
the 5hd2a system um i i talk about two regimes potentially um one is more towards um overt in
action and modifying the world and strengthen beliefs potentially of that variety and the other might be
more of um adapting to the world uh it's so uh robin has a a really excellent paper called a tale of two
receptors where he talks about 5hd2 5hd1a receptors the ones that antidepressants tend to act on um
because they tend to reduce this excitatory signaling um talking about them as a kind of passive coping
and so this would correspond in um there's some uh machine learning models of this by diane and
colleagues of like where they talk about opponents between dopamine and serotonin and they're focusing on
the 1a as a kind of like patience promoting parameter keeping you from like confidently charging into your
policy so that would tend to uh inhibit i guess patterns of um either overt inaction or potentially
even some kinds of um uh mental simulation too if it's like you could maybe even think of like even
if you're not overtly inactive enacting something externally uh there might be more or less uh brave
type styles of thoughts that you might be willing to entertain going um
uh going places you might not otherwise um be willing to go and being able to handle the swings
and expected free energy that might happen as you try to pursue a given path of policies
so but the idea would be that um there might be something more like a microdosing mesodosing
regime which might be more of what initially sculpted the 5hd2a system which evolved with the gene

duplication event around the time of uh the cambrian explosion with the advent of jawed fishes i'm personally interpreting this as having the significance of um evolving in the context of predator-prey arms races i suspect i could be wrong but that basically it's um locally trying to strengthen patterns of uh policy pursuit to pursue particular goals in a given context and this to come back to what i was watching earlier with i worked by robin of like a kind of active coping but it seems with rebus um there's almost like um not that kind of story um and that uh it's more you're adjusting the world and and actually if under a very ultra state you are overtly enacting that's actually kind of a dangerous thing i don't know how much to uh make of this that the um 5hd2a receptors are less expressed on like motor cortex i don't think we we know enough to make too much of it but it's like some dreams you might not want to act out as they're happening but they still might powerfully impact you so um i'm currently uh actually wondering whether this other uh rebus-like regime might have been specifically sculpted in the context of um bonding and actually opening the self to include others and that might have been some of the primary selective pressures i don't know how to think of that in terms of inaction um but uh and one more detail that might kind of like uh not sure what to make of it that kind of goes through both of those um excuse the phone uh would be um anthropologically i believe like psychedelics are oftentimes used in ways that would be both like creating um a kind of group solidarity and might be used in the context of um pursuing goals together in the world so for instance like a hunting party they might all do psychedelics together or something that looks a lot like breath work before uh going out maybe you know going to war so that's a very different type that kind of includes both a relaxation of the normal boundaries of the self and a kind of like uh gearing you up for pursuing goals in the world so but you know there's got to be another story of like like so arthur's expertise is machine learning and so it's like uh these you want these systems to act intelligently and not act in other times and so basically now we're in the language of policies and when you deploy them and how and it's so complicated yeah i would say that action manifests itself at a few different places it's true i didn't mention it here yeah i think it's very important obviously i mean i think what the kind of like most implicit place where it exists right is that um these belief landscapes are being instantiated ultimately for the sake of adaptive behavior adaptive behavior adaptive action right in some sort of environmental context and that's the kind of like framework deep learning framework that we're using or like the um artificial agent framework right is that like the assumption is that the system is supporting some sort of agent that's embodied in some sort of environment which is you know changing over time um such that it can either be adaptive or not adaptive i think then the question of how action actually plays into the evolution of these belief landscapes you know during psychedelics or not um you know acting is a acting in the world is a very clear way to um modify one's setting right so acting is a way of modifying the function that's providing the sensory input into the system which is going to evolve change how the activity patterns are evolving over time right which is going to affect the acute quality of the trip um affect how beliefs are either being strengthened or relaxed um and i think you know it's it's already clear that there's a kind of whole um set of like folk psychology right that's been built around like what is or isn't a good way to uh cultivate a helpful setting and that

of course all is a function of action in the world i think then there's also um this kind of more subtler form of action which is how one thinks how one deploys attention right essentially during this experience and i think this deployment of attention has a huge effect on whether there end up being rebus or cbus like effects right so whether one allows oneself to go down certain thought rabbit holes whether one allows oneself to get kind of stuck in certain kinds of thought patterns um i think this form of mental action yeah has a very big downstream effect um and i think a lot of i'd love to see much more work on this i think it's very important i think it's kind of essential um and i think this is kind of one of the most interesting things about these psychedelic experiences is that in some sense like the most extreme rebus model and the most extreme uh thebus model could be true for two different individuals right someone could say you know i went to hell and i was there for eight hours straight or um or say you know i had a completely selfless experience i like let go of all beliefs and experienced you know nothing for eight hours straight it was great right and like um clearly those involve like two very different neural regimes and i think the nature of the and you know obviously given that they involve the same drug and they involve not such a different setting it's obvious that they um something very important to do with how attention is deployed during this experience yeah yeah just uh i do think that this frame of attention i mean attention is so important for for everything and i do think that there's something interesting kind of digging into the the attention and albus uh framework that um yeah i'm really curious to see how that'll that'll develop i i would also say that um in terms of active inference and action uh there is a cost to holding an intention um and then you once you know you discharge your intention you can release
that cost um so i i guess i would have to think about how to apply that to deep canals framework
um um fred barrett um i don't know if he's published it yet but this is kind of like a follow-up to a
a claustral essential model that arthur alluded to uh um is emphasizing basically um a lack of executive control and placing that as central to the psychedelic spirits like not controlling your mental action in directed ways as much and then trying to come from more of the um types of cognition and phenomenology sort of psychedelics based on just releasing of uh the normal uh more agentic modes of policy selection around attention as mental action although he uh not using active inference speak so
yes yeah very interesting adam your um your introduction there of of course attention as covert action which was echoed by by arthur and michael both of you said it really introduces uh a case where we can use active inference to find continuity between what's happening inside the agent and outside the agent because on one hand we could kind of take a belief oriented frame and we could say well what's happening inside the agent are beliefs about things beliefs about hidden states beliefs about observations beliefs about policy so that's kind of a belief oriented view where everything is framed basically cognitively or we could take a very action oriented view and then we could think about the interface in a more inactive way and then also even the realm of where where many active entities uh dare not tread in mental action with attention as covert action and so then we have like a a total active account that is mathematically isomorphic with the total belief based account they're both just maps they're not the territory the territory is the the person in the room so all of this is just talking

about the formalism and how we choose to interpret the formalism in terms of it being more inferential
or it being more active um and so there's a lot of ways to go i guess in our closing minutes i would love to
hear what are the next moves or what are some areas that you um would hope that people look into
Michael and then the authors
okay so i know is um i think my my favorite mental move is to try to drop down on our level of
abstraction okay how how is this implemented um what's what's going on here and then i do think
that details of implementation um are really generative to look into and like the biomarkers and
proxies for you know figuring out how to how to parameterize okay exactly how hot is are we talking
about with this temperature parameter and and so on and in different parts of the body um i i have kind of
my my niche that i'm i'm working on in the vasomuscular system uh which is uh a rabbit hole but um
yeah i i just think that like i do think that this like the deep canals framework specifically
and the canalization paradigm in general um it's good enough that we should be able to dig into mechanism
so i'm i'm looking forward to that
great
great
adam
or arthur
go
so um or i think you can go first feeling doesn't matter
okay yeah okay sure um yeah i mean first i want to echo this idea uh i think this is very important
this is one of you know the people we've talked to about talked with about deep canals this idea of like
wow this is great you have these two axes you know it'd be wonderful if there are some like
biological core let's see if it look for um i think that's incredibly important um my background
is also in machine learning uh in deep learning so i'm currently doing some work around like
simulating some of these things i think uh especially for these albus dynamics uh
uh being able to kind of like study under what circumstances you've got these different kinds of
effects even in kind of like simpletory models is kind of important to see how they unfold um
and then i'm also you know we brought it up at the end i think it's really
important it's very interesting to me is the role of the deployment of attention in this um and there's
lines of work where people are trying you know various researchers are trying to connect for
example meditation to psychedelic research and kind of certain people are trying to tell unified stories
about um what's going on in the brain and people meditate what's going on with your psychedelics
um some of them using these active inference models um but i think in all of these cases the
deployment of attention and how the deployment of tension changes over time is kind of a key aspect
of this um and it's not as it's not as present in the deep canals model right now um but i think
any any really robust future kind of like model of these things is going to have to talk about attention
quite a bit
um i guess next steps would be um seeing how the review process goes um but also um

i guess uh helping arthur to establish a field of uh psychedelic machine learning psychedelic inspired machine learning where we use uh we try to make systems more uh flexible and capable better they can handle edge cases better imagine better in terms of real-time policy selection learn in ways that uh not too much not too little um and then maybe you know so that's useful potentially earlier like you have a driverless vehicle do i change lanes or don't i you send it back to the tesla dojo what do you want to learn what do you not want it to learn um but then down the line um
well before we even get there you know as you're going into this you could see the capacity for like a rich bi-directional flow between machine learning and um psychedelic inspired machine learning so it's like you get a bunch of machine learners together and a bunch of people who know a lot about the cognitive science and biology and maybe phenomenology of psychedelics and in addition to designing better machine learning systems they could probably tell you more about how psychedelics act and so you can have this kind of virtuous cycle between uh the biology and the field but you keep going with that
um i guess before we get there we'll keep going but before we get there um very interested in the implications for best practices so for instance like there might be like when so oftentimes you might want semantically neutral energy to kind of get them you're turning up the heat you're just maybe at first breaking up you know i call them defenses but the usual structure but then the question is at what point do you try to direct the canalization process do you wait till after or maybe when the when the metal is hot is that a really good time to do it and so these are the kinds and maybe there's no one single question this might depend on the substance the set the setting the dose the person so getting clarity on that could be uh faithful for many people in terms of um the outcomes they have and i specifically what i'm gunning for is i'm on a mission to make uh compassion meditation uh treatment as usual for uh all psychedelic psychotherapy and preferably all therapy and so the question is is
like if you're in a state of enhanced imaginative capacity maybe you're having more ability to tune into your interoceptive system whether you've just increased the channels or maybe you're better capable of because maybe your inflammation went down a bit and you can tune in better because it doesn't feel it feels better to do so some combination that might be a really good opportunity
to do certain meditations that might update you in particular ways like try to direct the re-annealing process maybe not just like afterwards integration but during the session and so if we could get clarity on something like um how to canalize and recanalize and the like annealing schedules like it's gonna be complicated but that's i think where we need to head and then finally um for all this i'm very uh
that would be enough but um recentering the conversation on psychedelics on basically
meaning and connection would be enough but i'm wondering uh down the line when we talk of alignment we're talking about our alignment with our values and maybe the alignments of machines with our values and so does this actually end up going to some of the conversations that are happening now about increasingly advanced ai and our relationships to it could there be a story of canalization and recanalization uh you want some things to hold fast and you want some things to be more fluid this might be it's always been a timely conversation we've always been dealing with this with us the alignment problem of us which we often fail but the alignment problem with these technologies we're developing

okay wow well so many great topics thank you all all i have in closing turn on tune in act and first serve

thank you thank you thank you thank you thank you thank you thank you thank you thank you thank you thank

Thank you.